

Multiple Choice (10 marks)

1. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following is a starvation-free policy?
 - (a) Youngest job first
 - (b) Least frequently used
 - (c) Round-robin
 - (d) Most recently used
 - (e) Longest job first
2. Let `l = [1,2,3,4]`. What is `min(l)` in Python3?
 - (a) '1'
 - (b) 0
 - (c) -1
 - (d) 3
 - (e) 1,2,3,4
3. Which is the largest asymptotically?
 - (a) $O(n^2)$
 - (b) $O(n^3)$
 - (c) $O(\sqrt{n})$
 - (d) $O(2^n)$
 - (e) $O(\log n)$
4. If the peak voltage value of a signal is 20 times the peak voltage value of the noise, what is the SNR? What is the `SNR_dB` (in 3 sig.fig.)?
 - (a) 15.0
 - (b) 13.0
 - (c) 30.0
 - (d) 34.0
 - (e) 26.0
5. Which of the following statements about a remote procedure call is true?
 - (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
 - (b) It cannot return a value.
 - (c) It cannot pass parameters by reference.
 - (d) It cannot call procedures implemented in a different language.

True / False (10 marks)

Answer True or False

6. True or False: A convex 3D object with 15 vertices and 39 edges has 20 faces, which is a common polyhedron.
7. True or False: In Python 3, the expression `b[::-2]` returns the elements at odd indices, resulting in a reversed list of every second element.
8. True or False: The boolean expression `a[i] == max || !(max != a[i])` simplifies to `a[i] <= max`, indicating if the element is less than or equal to the maximum.
9. True or False: A simple planar graph with 30 vertices can have a maximum of 84 edges according to the planar graph edge bound.
10. True or False: Clustering techniques are effective in detecting fraudulent credit card transactions by identifying outliers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check if the triangle is valid or not.
12. Write a function to sort the given array by using counting sort.

End of Paper